

RE026 - Multi-Factor Personalization in Food Recommendation Systems

Research Paper for Food Remedy API T1/2026

Prepared by: Ocean Ocean

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Purpose: This paper explains, in clear and practical language, how food recommendation systems combine multiple user factors to create safer, healthier, and more relevant food suggestions.

Executive Summary

Food recommendation systems are used in health, nutrition, recipe, grocery, and restaurant applications to suggest meals or food products that match a user. A basic recommender may only look at previous likes, ratings, or popular items. However, food is more complex than many other recommendation domains because the system must consider safety, health, culture, cost, time, and personal taste at the same time.

Multi-factor personalisation means that the recommendation engine combines several inputs before it recommends food. These inputs can include dietary goals, allergies, intolerances, cultural or religious preferences, health conditions, taste, budget, cooking time, location, and available ingredients. The aim is not just to recommend something the user might like, but to recommend something that is suitable, safe, realistic, and useful.

Research shows that modern food recommender systems commonly use hybrid approaches. A practical system may first filter out unsafe or unsuitable foods using strict rules, then rank the remaining options using a weighted score or machine-learning model. More advanced systems use multi-objective optimisation to balance competing goals such as taste, health, diversity, cost, and preparation effort. Studies also show that context-aware systems can improve relevance by considering time, place, and current situation.

For the Food Remedy API project, the key lesson is that recommendations should not be based on one input only. The system should treat allergies and medical restrictions as hard rules, then balance softer factors such as taste, nutrition goals, cost, and convenience. The app should also explain why a recommendation was made, because transparency improves trust and user satisfaction.

Ticket RE026 research task: Analyse how different applications incorporate multiple user inputs into recommendation systems. Study how conflicting factors, such as taste versus health goals, are prioritised. Compare approaches used across platforms and summarise how these systems manage complexity effectively.

1. Introduction

Food choices are personal. A meal can be healthy for one user but unsuitable for another because of allergies, medical conditions, religious requirements, cultural preferences, budget, or cooking time. This is why food recommendation systems need more than a simple “people also liked” approach.

A food recommender in a project such as Food Remedy API should support personalised nutrition by combining different types of information. For example, one user may want high-protein vegetarian meals, another may need low-sodium meals for blood pressure, while another may require gluten-free food because of coeliac disease. The system must understand these differences and generate suggestions that are practical and safe.

This report explains how multi-factor food recommendation works, what factors are commonly used, how systems combine these factors, and how they manage conflicts between competing goals. It also provides practical recommendations that can be applied to the Food Remedy API project.

Simple idea: A good food recommender should not only ask “Will the user like this?” It should also ask “Is it safe, healthy, culturally appropriate, affordable, and realistic for this user today?”

2. What Is Multi-Factor Personalisation?

Multi-factor personalisation is the process of using more than one user factor to decide what food to recommend. In a normal recommender, the system may focus mainly on past ratings or popularity. In a multi-factor recommender, the system combines several layers of information before presenting the final result.

In food recommendation, this matters because some factors are flexible while others are not. For example, a user’s preference for spicy food is flexible because they may still accept a mild meal. A peanut allergy is not flexible because the system must never recommend a food containing peanuts. Therefore, modern food recommenders often separate hard constraints from soft preferences.

Factor	Example user input	How the system should treat it
Allergies / intolerances	Peanuts, shellfish, lactose, gluten	Hard rule: exclude unsafe foods before ranking
Diet type	Vegetarian, vegan, halal, low-FODMAP	Usually hard rule or strong preference
Health condition	Diabetes, hypertension, kidney disease	Hard or strong rule depending on clinical risk
Nutrition goal	Weight loss, muscle gain, low sugar, high protein	Soft rule: rank foods that better match the goal
Taste preference	Spicy, mild, favourite cuisines, disliked ingredients	Soft preference: improve relevance and satisfaction
Context	Time of day, location, budget, cooking time	Soft factor: make recommendations practical now
Variety	Avoid repeating the same meal often	Soft factor: improve long-term adherence

3. Core Recommendation Techniques

- Content-based filtering:** Recommends foods similar to foods the user already likes, based on ingredients, nutrients, cuisine, or cooking style. Example: if the user likes high-protein chicken bowls, suggest similar high-protein meals.
- Collaborative filtering:** Uses behaviour from many users. Example: people with similar preferences saved this recipe, so it may also suit the current user.
- Knowledge-based or rule-based recommendation:** Uses explicit rules from nutrition, safety, or diet requirements. Example: if user has nut allergy, remove all recipes containing nuts.

- **Hybrid recommendation:** Combines multiple methods. A common approach is to use rules for safety, content-based features for food similarity, and collaborative filtering for taste prediction.
- **Context-aware recommendation:** Uses situation data such as time, location, weather, device, or current activity. Example: suggest a quick lunch near the user at 1 pm, but suggest home-cooked dinner recipes in the evening.

Research support: Food recommender system reviews describe content-based, collaborative, knowledge-based, hybrid, health-aware, and context-aware approaches as major directions in this field (Bondevik et al., 2024; Trattner & Elsweiler, 2017).

4. How Multiple User Inputs Are Combined

4.1 Step 1 - Collect User Inputs

A multi-factor system begins by collecting the user profile. This can be done through onboarding questions, account settings, food diaries, wearable data, past orders, saved recipes, and feedback buttons such as like, dislike, cooked, skipped, or not relevant.

- ☐ Dietary goal selected by the user
- ☐ Allergy and intolerance list
- ☐ Cultural or religious food rules
- ☐ Health condition or nutrition target
- ☐ Favourite cuisines and disliked ingredients
- ☐ Budget, cooking time, and preferred meal type
- ☐ Feedback on previous recommendations

4.2 Step 2 - Apply Hard Filters

Hard filters remove foods that should never be recommended. This step is important because it protects the user. Allergy, safety, and strict medical exclusions should be checked before any ranking occurs.

Example rule logic:

If user_allergies contains "peanut", remove all recipes where ingredients include peanut, peanut oil, peanut butter, satay sauce, or possible allergen tags.

4.3 Step 3 - Rank the Safe Options

After unsafe or unsuitable items are removed, the system can rank the remaining foods. A simple approach is to create a combined score from several factors:

$$\text{Final Score} = 40\% \text{ Taste} + 30\% \text{ Nutrition Fit} + 15\% \text{ Context Fit} + 10\% \text{ Cost Fit} + 5\% \text{ Variety}$$

The exact weights can change depending on the app. For a health-focused app, nutrition may receive a higher weight. For a lifestyle recipe app, taste and convenience may receive higher weights. Some systems also allow users to adjust these priorities using simple settings such as “more healthy”, “faster meals”, or “lower cost”.

4.4 Step 4 - Explain the Recommendation

Explanations help users trust the system. A useful food recommendation should not just show a recipe; it should briefly explain why the recipe is suitable.

- “Recommended because it matches your high-protein goal and avoids lactose.”
- “Good for your low-sodium plan and ready in under 20 minutes.”
- “Vegetarian, budget-friendly, and similar to meals you saved before.”

5. Managing Conflicts Between Factors

Conflict	Problem	Practical solution
Taste vs health	A user may prefer high-sugar desserts, but their health goal may require lower sugar.	Offer healthier swaps, smaller portions, or similar recipes with improved nutrition.
Convenience vs nutrition	A nutritious recipe may take one hour to cook, but the user may only have 15 minutes.	Recommend quick healthy meals on busy days and more complex meal-prep options on weekends.
Budget vs quality	A meal may be nutritionally ideal but too expensive.	Recommend cheaper ingredient alternatives or budget meal plans that still meet the main nutrition target.
Individual vs group needs	A family or group meal may need to satisfy several people with different restrictions.	Respect all allergies first, then balance shared taste preferences and rotate cuisines fairly.
Strict diet rules vs user freedom	If the system is too strict, users may feel controlled and stop using it.	Explain why rules matter and offer several acceptable choices, not only one fixed answer.

Research support: Studies on personalized meal planning and many-objective food recommendation show that balancing preference, nutrition, diversity, and user patterns can create more realistic and balanced recommendations than using preference alone (Zhang et al., 2022; Amiri et al., 2023).

6. Examples of Multi-Factor Food Recommendation Systems

- **Health-centric meal planner:** Uses user health goals, diet rules, calorie targets, and food preferences to generate daily or weekly meal plans. This is useful for users managing diabetes, obesity, hypertension, or general wellbeing goals.
- **Allergy-safe recommender:** Uses ingredient and allergen tags to filter unsafe foods before ranking. This is important because showing unsafe foods can damage trust and create real risk.
- **Context-aware restaurant recommender:** Uses the user’s location, time, health profile, and preferences to suggest suitable food nearby. Research prototypes have used mobile devices, beacons, medical records, and recommender engines to support real-time food decisions.
- **Nutrient-based recommender:** Suggests meals that match nutritional expectations and dietary restrictions while still considering user preference. The Yum-Me system is a well-known example of a personalised nutrient-based meal recommender.
- **Many-objective diet recommender:** Uses optimisation to balance multiple objectives, such as preference, nutrition, diversity, and diet patterns, instead of focusing on only one score.

7. Benefits for Recommendation Quality and User Satisfaction

Benefit	How it improves the app
More relevant suggestions	The app understands more of the user’s real situation, not only past likes.
Higher safety	Allergy and medical restrictions are checked before food is recommended.
Better trust	Users are more likely to trust recommendations when the system explains why they fit.
Better long-term use	Considering cost, effort, and variety makes recommendations more realistic.
More inclusive design	Cultural and religious food preferences make the app feel respectful and personalised.
Healthier choices	Nutrition goals can be built into the ranking so that healthier options are promoted without removing user choice.

8. Challenges and Limitations

- **Data collection burden:** If the app asks too many questions at sign-up, users may abandon it. The app should collect the most important safety information first and allow users to add more detail later.
- **Data quality:** Ingredient lists, allergen tags, and nutrition values must be accurate. Incorrect data can lead to poor or unsafe recommendations.
- **Privacy:** Health, diet, allergies, and cultural preferences can be sensitive. The system should collect only required data, protect it securely, and explain how it is used.
- **Cold start problem:** New users have little history. The app should use onboarding questions and safe defaults until it learns more from feedback.
- **Explainability:** Complex machine-learning models may be hard to understand. Simple explanations should be added so users know why a food was recommended or removed.
- **Over-personalisation:** If recommendations become too narrow, users may get bored. Variety and discovery should be included.

9. Application to Food Remedy API

For the Food Remedy API project, multi-factor personalisation should be designed as a layered recommendation process. The system should first protect the user, then personalise for health, taste, and context.

Food Remedy component	What it should do
1. User profile	Store allergies, diet type, goals, cultural preferences, disliked ingredients, and optional health conditions.
2. Food database	Store ingredients, allergens, nutrients, cuisine, tags, preparation time, price estimate, and serving size.
3. Safety filter	Remove foods that conflict with allergies, strict diet rules, or high-risk medical restrictions.
4. Scoring engine	Rank remaining foods using taste, nutrition fit, context fit, cost, effort, and variety.
5. Explanation layer	Show a short reason: “matches your low-sugar goal”, “gluten-free”, “quick dinner”, or “similar to meals you liked”.
6. Feedback loop	Allow the user to like, dislike, save, hide, or report recommendations so the model improves over time.

10. Interactive Checklist for the Project

This checklist can be used by the Food Remedy API team to confirm that multi-factor personalisation has been considered properly.

- ☐ Can the user add allergies and intolerances?
- ☐ Are allergy rules treated as hard filters before ranking?
- ☐ Can the user choose diet type such as vegetarian, vegan, halal, gluten-free, or low-FODMAP?
- ☐ Can the user set goals such as weight loss, muscle gain, low sugar, high protein, or low sodium?
- ☐ Does the food database include ingredients, allergens, nutrients, cuisine, and preparation time?
- ☐ Does the recommendation score combine taste, health, context, cost, and variety?
- ☐ Does the system avoid repeating the same meals too often?
- ☐ Does the app explain why each recommendation was made?
- ☐ Can the user give feedback to improve future suggestions?
- ☐ Are sensitive health and dietary data handled with privacy and security controls?

11. Recommendation Priority Model

A simple priority model for Food Remedy API is shown below. This model keeps safety first while still allowing personalisation.

1. **Priority 1 - Safety:** Allergies, intolerances, and dangerous medical restrictions must be checked first. Unsafe foods are removed completely.
2. **Priority 2 - Required diet rules:** Vegan, halal, gluten-free, or other strict diet requirements should be strongly enforced.
3. **Priority 3 - Health and nutrition goals:** Rank foods higher when they match calorie, macro, sugar, sodium, fibre, or protein targets.
4. **Priority 4 - Taste and culture:** Improve satisfaction by matching cuisine, flavour, and past likes.
5. **Priority 5 - Context and practicality:** Prefer foods that match the user's current time, budget, location, cooking equipment, and available time.
6. **Priority 6 - Variety and discovery:** Avoid repetitive plans and occasionally include new but suitable foods.

12. Conclusion

Multi-factor personalisation is essential in food recommendation systems because food choices depend on many connected factors. A useful system must consider health goals, allergies, culture, taste, context, cost, and effort rather than relying only on popularity or past behaviour.

The best approach is to separate strict rules from flexible preferences. Allergy and safety rules should be handled as hard filters, while taste, nutrition goals, cost, and convenience can be balanced through scoring or optimisation. Hybrid systems are especially suitable because they can combine rule-based safety, content-based food matching, user feedback, and machine-learning predictions.

For the Food Remedy API project, the recommendation engine should be designed around safety, transparency, and practical usefulness. The app should provide recommendations that are safe, explainable, personalised, and realistic for everyday users. This will improve recommendation quality, user satisfaction, and trust in the platform.

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